Literature Review: Faithfulness and Retrieval Robustness in Retrieval-Augmented Generation

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Retrieval-Augmented Generation (RAG) grounds language model outputs in external documents to reduce hallucination without retraining (Lewis et al., 2020). Research has since branched in three directions: improving what gets retrieved, verifying whether generation stays faithful to it, and evaluating these systems at scale.

0.1 Retrieval Methods and Robustness

Dense passage retrieval (Karpukhin et al., 2020) replaced sparse methods with learned embeddings for semantic matching. However, retrieval quality remains fragile: Shi et al. (2023) showed LLMs degrade when irrelevant context is mixed with relevant passages, and Liu et al. (2024) demonstrated a positional bias where models neglect information in the middle of long contexts. On the representation side, ColBERTv2 (Santhanam et al., 2022) introduced late-interaction retrieval with token-level similarity, achieving stronger matching than single-vector bi-encoders.

Several approaches tackle retrieval quality directly. Corrective RAG (Yan et al., 2024) classifies retrieved documents as correct, ambiguous, or incorrect and triggers query refinement for low-confidence retrievals. Adaptive-RAG (Jeong et al., 2024) routes queries by complexity, while FLARE (Jiang et al., 2023) interleaves generation with retrieval by detecting low-confidence spans mid-generation. These reduce noise fed to the generator, but none measures whether this translates to more faithful citations.

0.2 Faithfulness and Self-Correction

Self-RAG (Asai et al., 2024) trains the LLM to emit reflection tokens signaling retrieval need, passage relevance, and generation support—internalizing the verify loop but requiring fine-tuning. RAFT (Zhang et al., 2024) fine-tunes on relevant and distractor documents, teaching models to cite correctly. RankRAG (Yu et al., 2024) unifies reranking and generation in a single instruction-tuned LLM.

For evaluation, FActScore (Min et al., 2023) decomposes generations into atomic facts verified against a knowledge source, shifting evaluation from surface overlap toward entailment-based verification. ALCE (Gao et al., 2023) extends this to citation-specific evaluation, benchmarking whether each generated statement is backed by a correct inline citation.

0.3 Evaluation Frameworks and Structured Retrieval

RAGAS (Es et al., 2024) defines component-wise metrics: faithfulness, answer relevancy, context precision, and context recall. ARES (Saad-Falcon et al., 2024) automates evaluation via LLM judges with statistical guarantees. The RGB benchmark (Chen et al., 2024) tests noise robustness, negative rejection, information integration, and counterfactual robustness—axes that overlap with but do not cover citation accuracy. Beyond flat vector search, HippoRAG (Gutiérrez et al., 2024) uses knowledge graphs with PageRank traversal, MemoRAG (Qian et al., 2024) introduces a dual-system memory architecture, and RECOMP (Xu et al., 2024) compresses passages before generation. Gao et al. (2024) survey these approaches, identifying retrieval granularity, noise filtering, and faithful generation as persistent open problems.

0.4 Positioning Our Work

We combine corrective retrieval (Yan et al., 2024) with post-hoc citation verification inspired by FActScore and ALCE, without fine-tuning the generator (unlike Self-RAG and RAFT). We move beyond general-domain benchmarks by constructing a domain-specific evaluation set for scientific policy literature. No prior work combines corrective retrieval, claim-level faithfulness scoring, and domain-specific evaluation in a single pipeline.

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